

Important Topics BCA Sem 1

BCA 1st Semester — Important & Repeatedly Asked Topics

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About These Tables

This file brings together important topics for BCA 1st Semester revision from two sources:

- **From Previous Year Question Papers** — topics compiled from combined question banks
- **From Personal Study Notes** — Quick Reference — Exam Essentials and Block-wise Revision Checklists (based on SGOU SLM)

Priority Markers

Use these symbols while revising to spot the most important topics.

★ = Asked in multiple years (High Priority) — *Previous Year Question Papers tables only*

🔥 = Very important topic — *must revise before exam*

Environmental Studies (B21ES01AC)

From Previous Year Question Papers

Topics in the table below are compiled from past exam question papers.

Topic	Asked In	Sections	Priority
Sustainable Development (definition, SDGs, sustainable food system)	Aug 2023, May 2024, Apr 2025	A, B, C, D	★🔥
Climate Change & Global Warming (causes, consequences, mitigation,	Aug 2023, May 2024, Apr 2025	A, B, C, D	★🔥

Topic	Asked In	Sections	Priority
impacts, greenhouse gases, carbon footprint)			
Solid Waste Management (3Rs, disposal methods, Zero Waste, strategies, definition)	Aug 2023, May 2024, Apr 2025	A, B, C, D	★🔥
Swachh Bharat Abhiyan (note, objectives, waste cleanliness)	Aug 2023, May 2024	B, C	🔥
Occupational Health & Disease (occupational disease, health hazards, environmental diseases)	Aug 2023, May 2024, Apr 2025	A, B, C, D	★🔥
Environmental Impact Assessment (EIA) (importance, aim, definition, stages)	Aug 2023, May 2024, Apr 2025	A, B, C	★🔥
Weather vs Climate	Aug 2023, May 2024, Apr 2025	B, C	★🔥
Anthropocentrism & Environmental Ethics (scope, definition)	Aug 2023, May 2024, Apr 2025	A, B	★🔥
Ecosystem (food chain, food web, ecological pyramid, biotic/abiotic components, terrestrial/aquatic types)	Aug 2023, May 2024, Apr 2025	A, B, C	★🔥
Biodiversity & Conservation (in-situ, ex-situ, hotspots, flagship species, values, Sundalands, strategies)	Aug 2023, May 2024, Apr 2025	A, B, C, D	★🔥
Pollution (water pollution, Air Act, ozone depletion, microplastics, acid rain)	Aug 2023, May 2024, Apr 2025	A, B, C	🔥
Natural Resources (renewable vs non-renewable, conservation methods, biogas, threats)	Aug 2023, May 2024, Apr 2025	A, B, C	★🔥
Atmosphere (layers, troposphere, ozone layer, hydrological cycle, water cycle)	Aug 2023, May 2024, Apr 2025	A, B, C	★

Topic	Asked In	Sections	Priority
Environmental Movements (Chipko, Narmada Bachao Andolan, green movement)	May 2024, Apr 2025	A, B, C	★🔥
Vector Borne Diseases (Malaria vector, disease account)	May 2024, Apr 2025	A, C	★🔥
Population & Its Effects (growth issues, overpopulation essay, population explosion)	Aug 2023, May 2024, Apr 2025	A, B, C, D	★🔥
Wildlife Protection Act / CITES / CRZ (acts, abbreviations, salient features, sanctuaries)	Aug 2023, May 2024, Apr 2025	A	★
Natural Disasters (earthquakes, landslides, floods, storm surge)	Aug 2023, May 2024, Apr 2025	A, C	★
Eco-club & Green Economy (Haritha Keralam Mission)	Aug 2023, May 2024, Apr 2025	B, C	★
Carbon Sequestration & Climate Management	Aug 2023, May 2024	C	🔥
Forests in India (types, classification, dense forest rainfall)	May 2024, Apr 2025	A, D	★🔥
Environmental Economics & Equity (green economy, environmental justice)	Aug 2023, Apr 2025	B, C, D	★
Soil Erosion & Desertification (causes, control methods)	Aug 2023, Apr 2025	B	★
Water Resources (scarcity, threats, groundwater, ocean percentage)	Aug 2023, May 2024, Apr 2025	A, C	★

From Personal Study Notes

My personal study notes based on SGOU SLM — Environmental Studies (B21ES01AC).

Quick Reference — Exam Essentials

Topic	Must-Know
Four segments 🔥	Atmosphere, Hydrosphere, Lithosphere, Biosphere
Atmosphere 🔥	N ₂ 78%, O ₂ 21%; troposphere = weather zone
Water on Earth 🔥	71% hydrosphere; hydrological cycle
Forest types 🔥	Tropical moist/dry, montane, alpine (16 types)
Ecosystem 🔥	Producers → Consumers → Decomposers
Food chain/web 🔥	Linear vs interconnected feeding relationships
Ecological pyramid 🔥	Numbers, biomass, energy (energy always upright)
Biodiversity 🔥	Genetic, species, ecosystem levels
India hotspots 🔥	Western Ghats, Himalaya, Indo-Burma, Sundaland
Protected areas 🔥	NP (strict), Sanctuary (habitat), Biosphere Reserve (UNESCO)
Threats	Habitat loss, poaching, man-wildlife conflict
SD definition 🔥	Brundtland 1987 — needs of present without compromising future
GHGs 🔥	CO ₂ , CH ₄ , N ₂ O, CFCs — greenhouse effect
Ozone 🔥	Stratosphere; depleted by CFCs
Carbon credit	Tradeable permit to emit CO ₂ ; cap and trade
Acts 🔥	Wildlife 1972, Water Act, Air Act, Environment Protection Act
Movements 🔥	Chipko (trees), Narmada Bachao (dam)
Waste 🔥	Reduce, Reuse, Recycle; 3-bin segregation

Block-wise Revision Checklist

Block	Key Topics to Revise
Block 1 — Environment & Natural Resources	Four segments; atmosphere composition & layers; hydrological cycle; weather vs climate; renewable vs non-renewable resources; forest types (16 types), social forestry & agroforestry; water distribution, floods, drought, CRZ; land degradation, soil erosion, desertification; energy types (fossil, solar, wind, biogas, nuclear)
Block 2 — Ecosystems & Biodiversity	Abiotic & biotic components; food chain, food web, ecological pyramids; terrestrial & aquatic ecosystems; biodiversity levels (genetic, species, ecosystem); six values of biodiversity; India as mega diversity nation; four hotspots; threats & Red Data Book; in-

Block	Key Topics to Revise
	situ (NP, Sanctuary, Biosphere Reserve) vs ex-situ conservation; IUCN categories
Block 3 — Social Issues & Sustainable Development	Environment & human health; occupational diseases (Asbestosis, Silicosis, Fluorosis, Asthma); water-borne, vector-borne & air-borne diseases; public awareness; Chipko Movement, Narmada Bachao Andolan; Swachh Bharat, Haritha Keralam, Eco-club; population growth & explosion; Brundtland definition (1987); SDGs
Block 4 — Ethics & Contemporary Issues	Environmental ethics (Anthropocentrism, Biocentrism, Ecocentrism, Ecofeminism); solid waste management, 3-bin segregation, zero waste; greenhouse effect, global warming, climate change; acid rain, ozone depletion; carbon sequestration, carbon footprint, carbon credits; green economy, circular economy, EIA; landslides, floods, cyclones, earthquakes; Wildlife Protection Act 1972, Air Act, Water Act, Environment Protection Act

Reading and Writing English (B21EG01LC)

From Previous Year Question Papers

Topics in the table below are compiled from past exam question papers.

Topic	Asked In	Sections	Priority
The Blessed Damozel (symbolism, themes, lily, love beyond death)	Aug 2023, May 2024 (×2), Apr 2025	A, B, C	★🔥
The Open Window (Vera, prank, window symbol, atmosphere)	Aug 2023, May 2024, Apr 2025, Oct 2025	A, B, C	★🔥
Ignited Minds / The Dream and the Message (Kalam, thinkers, youth)	May 2024 (×2), Apr 2025	A, B, C, D	★🔥
Communication (forms, barriers, encoding/decoding, effective)	All 5 exams	A, B, C, D	★🔥
Non-verbal Communication (elements, paralanguage, proxemics)	May 2024, Oct 2025	B, C, D	🔥

Topic	Asked In	Sections	Priority
Summary vs Precis vs Abstract	Aug 2023, May 2024, Apr 2025, Oct 2025	A, B, C	★🔥
Skimming vs Scanning	May 2024, Oct 2025	B, C	★🔥
Types of Reading (categories, significance)	Aug 2023, May 2024, Apr 2025, Oct 2025	B, C, D	★🔥
Report Writing (types, magazine report, progress report)	Aug 2023, May 2024 (×2), Apr 2025, Oct 2025	B, C, D	★🔥
Email Writing (responses, acknowledgments, business reply)	May 2024 (×2), Oct 2025	A, C, D	★
Editing & Proofreading (steps, benefits, error correction)	Aug 2023, May 2024, Apr 2025	B, C	🔥
Active vs Passive Vocabulary	Aug 2023, May 2024 (×2)	B	★
Grammar — Voice (active ↔ passive conversion)	Aug 2023, Apr 2025	A, B, C	★
Grammar — Concord (subject-verb agreement)	Aug 2023, May 2024, Oct 2025	A, C	★🔥
Grammar — Sentence Types (simple, compound, complex)	Aug 2023, May 2024, Apr 2025	B, C, D	★🔥
Grammar — Tenses (future, continuous, reported speech)	Aug 2023, May 2024, Apr 2025, Oct 2025	A, B, C	★
Parts of Speech / Adjectives / Determiners	May 2024, Apr 2025, Oct 2025	C, D	★
Reading Comprehension (passage-based, close reading)	May 2024, Apr 2025, Oct 2025	B, C	★
Essay Writing (structure, argumentative, topics)	May 2024, Apr 2025, Oct 2025	A, B, D	🔥
Poem Appreciation (Stopping by Woods, Fire)	Aug 2023, May 2024	D	★

Topic	Asked In	Sections	Priority
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From Personal Study Notes

My personal study notes based on SGOU SLM — Reading and Writing English (B21EG01LC).

Quick Reference — Exam Essentials

Topic	Must-Know
Communication origin	Latin <i>communicare</i> = to share
Basic model 🔥	Sender → Message → Receiver → Feedback
Best style	Assertive
Informal network	Grapevine communication
Non-verbal studies 🔥	Haptics (touch), proxemics (space), chronemics (time), vocalics (paralanguage)
Barriers 🔥	Physical, psychological, linguistic, cultural, organisational, etc.
Reading types 🔥	Skimming, scanning, close reading, amusive, reading between lines
Close reading 🔥	3 steps; John Ciardi
Textbase levels	Linguistic, microstructure, macrostructure
Situation model	Text + reader knowledge → deepened understanding
Kalam essay 🔥	<i>Ignited Minds</i> ; dream 30 Sept 2001; five thinkers
Saki story 🔥	<i>The Open Window</i> ; Vera; window = symbol; childhood
Rossetti poem 🔥	<i>The Blessed Damozel</i> ; white lily = resurrection; love, grief, religion
Parts of speech 🔥	8: noun, verb, adj, adv, pron, prep, conj, interj
Articles	a (consonant sound), an (vowel sound), the (specific), zero (uncountable)
Sentence types 🔥	Assertive, interrogative, imperative, exclamatory
Question tags 🔥	Positive stmt → negative tag; I am → aren't I?; Let us → shall we?
Concord 🔥	Subject-verb agreement; 3 types; a number of vs the number of
Tenses 🔥	12 forms; non-conclusive verbs (know, believe, want)
Reported speech 🔥	Present → past; will → would; statement order for questions

Topic	Must-Know
Active/Passive 🔥	Object of active → subject of passive; be + past participle
Writing process 🔥	Plan, organise, draft, edit, revise
Summary vs précis 🔥	Summary = nutshell, no heading; précis = crux + heading, ~1/3
Abstract	≤300 words; keywords; self-contained
Father of Essays	Montaigne
Essay structure 🔥	Introduction, Body, Conclusion
Report 🔥	Who, what, when, where, why
Informal reports	Letter report, memo report
Proofreading 🔥	Final step; surface-level errors
Editing 🔥	Linking devices; ESL benefit

Block-wise Revision Checklist

Block	Key Topics to Revise
Block 1 — Glimpses of Communication	Communication model; sender, message, receiver, feedback; one-way vs two-way; verbal & non-verbal; encoding/decoding; communication styles (assertive best); barriers; active listening; haptics, proxemics, chronemics, paralanguage
Block 2 — Reading	Skimming, scanning, close reading; comprehension levels (linguistic, microstructure, macrostructure); textbase & situation model; <i>Ignited Minds</i> (Kalam); <i>The Open Window</i> (Vera, Saki); <i>The Blessed Damsel</i> (Rossetti, white lily)
Block 3 — Language Development	8 parts of speech; noun & verb types; articles; adjective degrees; sentence types (simple, compound, complex); question tags; concord rules; 12 tenses; reported speech; active/passive voice
Block 4 — Basics of Writing	Creative vs non-creative writing; content, form, structure, style; 5-step writing process; email responses (acknowledge, approve, decline); active vs passive vocabulary
Block 5 — Summary, Essays & Reports	Summary vs précis vs abstract; essay types & structure (Montaigne); report (5 wh- questions); informal vs formal reports; editing vs proofreading
Block 6 — Practice	Comprehension (main idea, inference); grammar error spotting; précis rules (heading, 1/3 length); essay (300 words); report (memo format)

Problem Solving and Programming in C (B21CA02DC)

From Previous Year Question Papers

Topics in the table below are compiled from past exam question papers.

Topic	Asked In	Sections	Priority
Algorithm & Flowchart (definition, symbols, problem-solving steps, draw flowchart)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Operators in C (priority, types, conditional operator, examples)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Loops (for, while, do-while, entry/exit controlled, loop control structures)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Call by Value & Call by Reference (default method, swap program, differentiate)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Recursion (definition, types, advantages/disadvantages, vs iteration, example program)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Functions (user-defined, advantages, odd/even program, return type, main)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Arrays (declaration, 1D array, sort, largest/second largest, average program)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Pointers (& and *, dereferencing, pointer address)	Dec 2024, Apr 2025	A, B	★🔥
Structures & Unions (definition, advantages/disadvantages, compare with array, member access)	Dec 2024, Apr 2025, Oct 2025	A, B, C	★🔥
Storage Classes (auto, extern, static, register — default, scope, initial values)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
File Handling (fopen, modes, append, fread/fwrite, rewind, file	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥

Topic	Asked In	Sections	Priority
pointer, file copy)			
Command Line Arguments (argc, argv, significance, file copy program)	Dec 2024, Apr 2025, Oct 2025	A, B, D	★🔥
Preprocessor & Macros (directives, #define, macros vs functions)	Apr 2025, Oct 2025	A, B, C	★🔥
Dynamic Memory (malloc, calloc, realloc, free)	Dec 2024, Apr 2025	B, C	★🔥
printf() & scanf() (header file, format specifiers, short notes)	Dec 2024, Apr 2025, Oct 2025	A, B	★🔥
C Tokens (types, identifiers, keywords, constants, symbolic names)	Dec 2024, Apr 2025, Oct 2025	B, C	★🔥
Control Structures (if-else-if ladder, switch-case, break, continue, goto)	Apr 2025, Oct 2025	B, C, D	★🔥
Type Conversion & Casting (methods, compare with type casting)	Dec 2024, Oct 2025	A, B, C	★🔥
String Handling (delimiter, strlen, string manipulation functions)	Dec 2024, Apr 2025	A, C	★
Output Prediction (for loop trap, printf format, register, infinite recursion)	Dec 2024, Apr 2025, Oct 2025	A, B	🔥
Structure of C Program (sections, symbolic constants)	Oct 2025	C	★

From Personal Study Notes

My personal study notes based on SGOU SLM — Problem Solving and Programming in C (B21CA02DC).

Quick Reference — Exam Essentials

Topic	Must-Know
Problem-solving 🔥	Analyse → Design → Code → Test

Topic	Must-Know
Algorithm 🔥	Finite, unambiguous; input, process, output
Flowchart 🔥	Oval=start/stop, rectangle=process, diamond=decision
Translators	Compiler (whole program), interpreter (line by line)
C basics 🔥	main(), semicolon, case-sensitive, #include
Data types 🔥	int, char, float, double, void
Operators 🔥	9/2=4; % remainder; ++a vs a++
scanf/printf 🔥	& required in scanf for variables
Control 🔥	if, switch (break!), for, while, do-while
Arrays 🔥	Index 0 to n-1; strings end with \0
Pointers 🔥	& address, * value; malloc/free
Functions 🔥	Prototype, definition, call
Recursion 🔥	Base case mandatory
Parameters 🔥	Call by value default; pointers for reference
struct vs union 🔥	All members vs shared memory
Storage class 🔥	auto, static, register, extern
Files 🔥	FILE *fp; fopen modes; fclose; EOF=-1
argc/argv 🔥	argc includes name; argv[0]=program
Preprocessor 🔥	#define, #include; runs before compiler

Block-wise Revision Checklist

Block	Key Topics to Revise
Block 1 — Basic Programming Concepts	Problem-solving steps; algorithm characteristics; flowchart symbols; C history (Dennis Ritchie); program structure; tokens & keywords; data types & modifiers; operators; type conversion
Block 2 — I/O, Control, Arrays & Pointers	printf/scanf & format specifiers; if-else, switch, loops (for/while/do-while); break & continue; 1D/2D arrays; strings & string functions; pointers (&, *); malloc, calloc, realloc, free
Block 3 — Functions, Structures & Union	User-defined functions; recursion (base case); call by value vs call by reference; local vs global variables; structures (. and →); structure vs union; typedef & enum
Block 4 — Storage Classes, Files &	auto, register, static, extern (scope & lifetime); fopen/fclose modes; fprintf, fscanf; fseek, rewind; argc & argv; #define

Block	Key Topics to Revise
Preprocessors	macros; macros vs functions

Introduction to Information Technology (B21CA01DC)

From Previous Year Question Papers

Topics in the table below are compiled from past exam question papers.

Topic	Asked In	Sections	Priority
Operating Systems (types, functions, classifications, mobile OS, NOS, DOS)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
System vs Application Software (definition, compare, utility, layers)	Dec 2024, Apr 2025	B, C	★🔥
HTML (elements, tags, links, lists, frames, media, documents)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Computer Networks (types, topologies, devices, WAN, hub/switch, Wi-Fi)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
CPU & Architecture (registers, accumulator, CISC/RISC, Von Neumann, instruction cycle)	Dec 2024, Apr 2025, Oct 2025	A, B, C, D	★🔥
Memory (DRAM, EEPROM, cache, virtual memory, nibble, hit ratio)	Dec 2024, Apr 2025, Oct 2025	A, B, C	★🔥
Security (malware, virus, antivirus, computer security)	Dec 2024, Apr 2025, Oct 2025	A, C, D	★🔥
Language Translators (interpreter, compiler, translator, high/low level languages)	Dec 2024, Apr 2025, Oct 2025	A, B, C	★🔥
Input/Output Devices (scanners, printers, hardcopy, USB)	Dec 2024, Apr 2025, Oct 2025	A, B	★
Number Systems (Gray code, ASCII, binary concepts)	Oct 2025, Dec 2024	A, D	★🔥

Topic	Asked In	Sections	Priority
Logical vs Physical Address	Dec 2024, Oct 2025	B, C	★🔥
URL & Web (structure, web pages, browsers, search engines, 404 error)	Dec 2024, Apr 2025, Oct 2025	A, B, C	★🔥
Emerging Technologies (AR, VR, cognitive computing, e-commerce, CAD)	Oct 2025	B, C, D	🔥
Office & Productivity Software (word processing, spreadsheets, presentation, LaTeX, email)	Apr 2025, Oct 2025	A, B, C	★
Computer Classification (hybrid, classifications, characteristics)	Dec 2024, Oct 2025	A, D	★🔥
Bootting (warm boot, POST)	Dec 2024	A, C	★
Metadata	Apr 2025, Oct 2025	A, B	★

From Personal Study Notes

My personal study notes based on SGOU SLM — Introduction to Information Technology (B21CA01DC).

Quick Reference — Exam Essentials

Topic	Must-Know
Computer functions 🔥	Input, Processing, Output, Storage
Hardware/Software/Firmware/Liveware	Physical parts / Programs / ROM (BIOS) / Users
Von-Neumann 🔥	Stored program; CU, ALU, memory, I/O
Number systems 🔥	Binary, Decimal, Octal, Hex; BCD, Gray, Excess-3, ASCII
Memory hierarchy 🔥	Registers → Cache → RAM → Secondary
Instruction cycle 🔥	Fetch → Decode → Execute → Store
RISC vs CISC 🔥	Simple/fast vs Complex/compact

Topic	Must-Know
Boot process 🔥	POST → Boot loader → OS (warm vs cold boot)
Translators 🔥	Compiler, Interpreter, Assembler
LAN/MAN/WAN 🔥	Building / City / Worldwide
Topologies 🔥	Bus, Ring, Star, Mesh, Tree, Hybrid
Email 🔥	SMTP send; POP3/IMAP receive
HTML skeleton 🔥	<code><!DOCTYPE html><html><head><body></code>
Banking IT	ECS, MICR, NEFT, RTGS, CBS, ATM
Security (CIA) 🔥	Confidentiality, Integrity, Availability
Emerging tech 🔥	VR, AR, AI, IoT, Quantum

Block-wise Revision Checklist

Block	Key Topics to Revise
Block 1 — Computer Fundamentals	Input, processing, output, storage; hardware, software, firmware, liveware; Von-Neumann architecture; number systems (binary, octal, hex); bit, nibble, byte; BCD, Gray code, ASCII; input/output devices; computer types (analog, digital, hybrid)
Block 2 — Instructions, Memory & Storage	Memory hierarchy; RAM vs ROM; SRAM vs DRAM; cache (L1/L2/L3, hit ratio); instruction cycle; RISC vs CISC; CPU registers (MAR, MDR, PC, IR, Accumulator); virtual memory; secondary storage (HDD, SSD, optical)
Block 3 — Software	System vs application software; OS functions; boot process (POST, warm vs cold boot); software layers; OS types (batch, time-sharing, real-time, distributed, network, mobile); GUI; compiler vs interpreter; DBMS; utility software; virus types & protection
Block 4 — Networks & Internet	LAN, MAN, WAN; hub vs switch vs router; repeater & bridge; network topologies; URL structure; search engines; ISP; Wi-Fi vs WiMAX; DSL; email (SMTP, POP3, IMAP); web pages & applications
Block 5 — HTML	HTML structure (DOCTYPE, head, body); document types; lists (ul, ol, dl); tables; frames; images & media; internal & external links; website creation steps
Block 6 — Trends in IT	IT in business, banking, education, medicine; mobile apps; bioinformatics; cognitive computing; AI types; IoT; cloud computing; security (CIA triad); malware & antivirus; VR & AR

Wishing you all success in your semester exams.

Dua mein Yaad Rakhna.